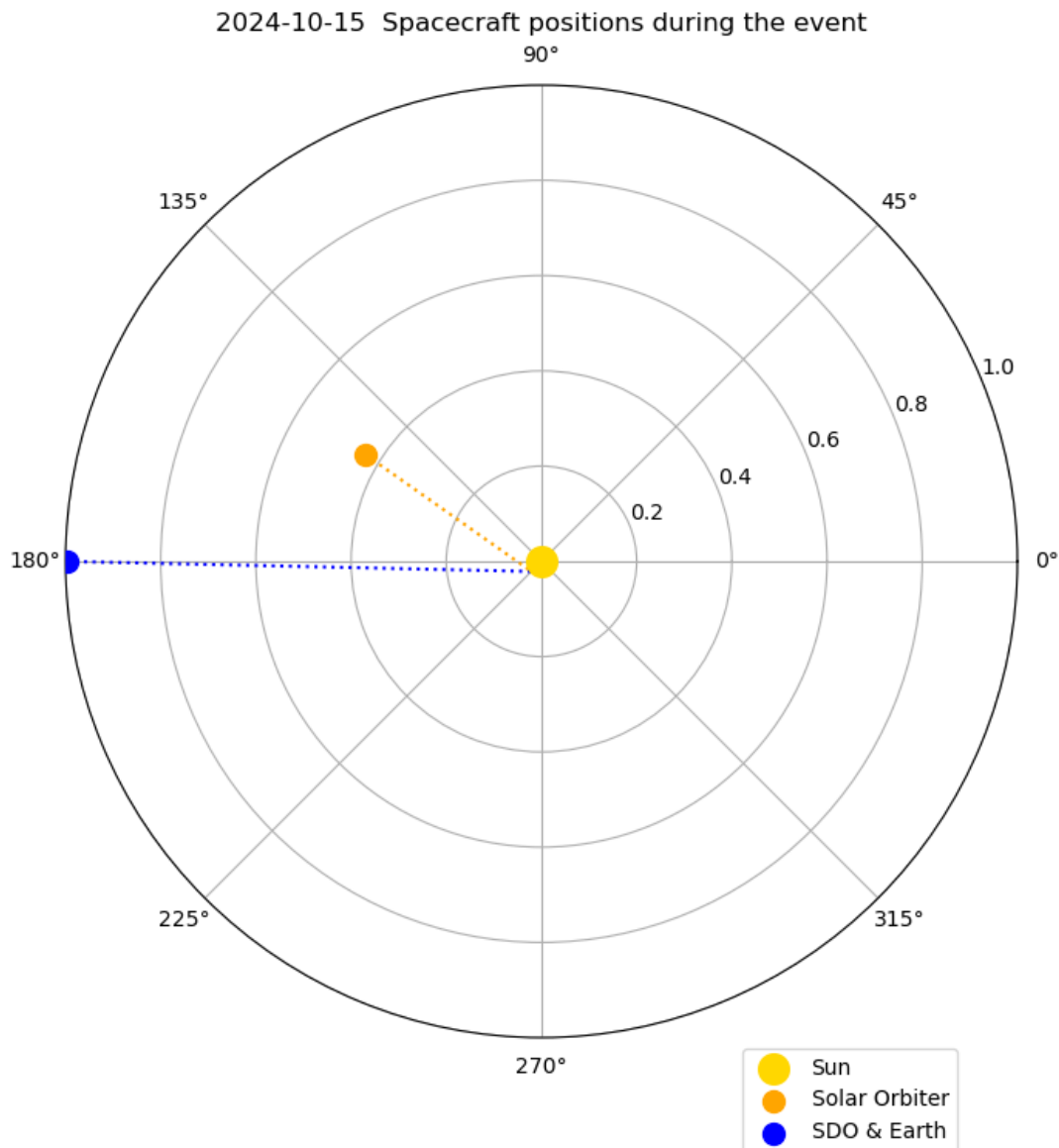


Summer student internship: Multiwavelength Analysis of Quasi-Periodic Pulsations (QPPs) in a Solar Flare | PROGRESS REPORT no. 3 | 19/08/2025

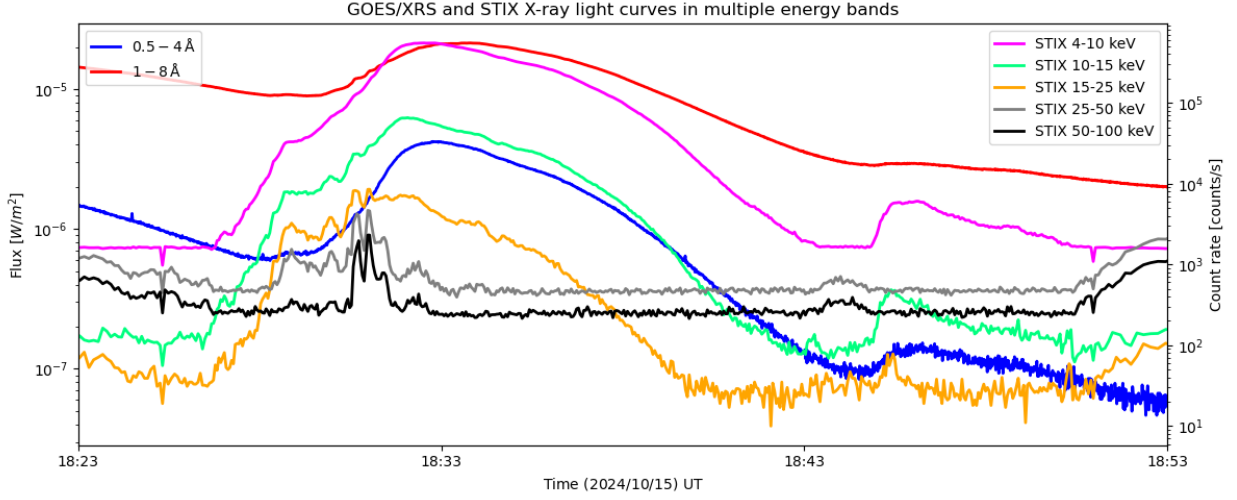
For the third progress report we continued with the analysis of QPPs in M2.0-class solar flare (occured on 15/10/2024 around 18 : 30 UT), which was observed during the Solar Orbiter Major Flare Watch Campaign. Analysis was done through multiple steps, and here we're showing key plots and conclusions.

Bellow, we show spacecraft positions when this solar flare occurred¹.

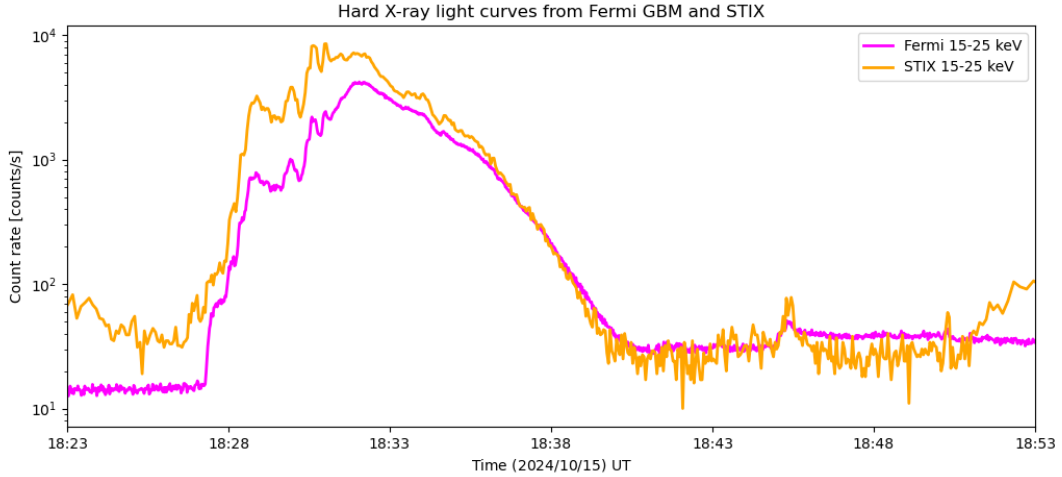


¹STIX has a slightly different viewing angle (31.2°); light travel time needs to be accounted for too.

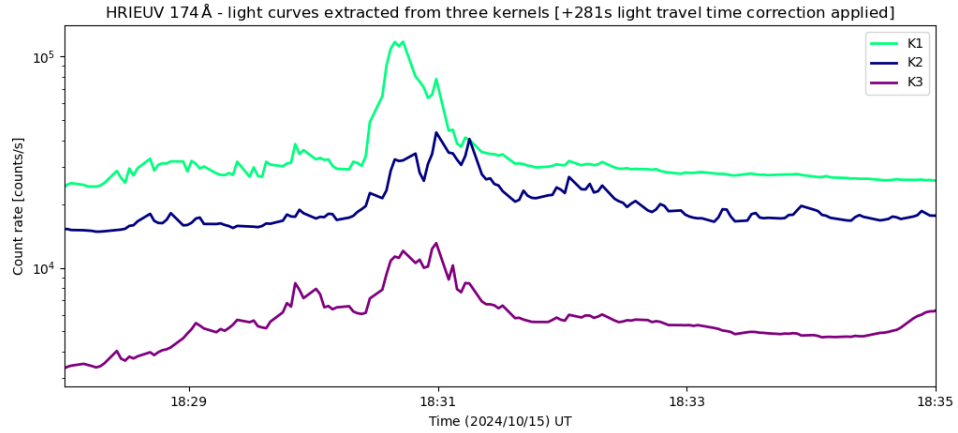
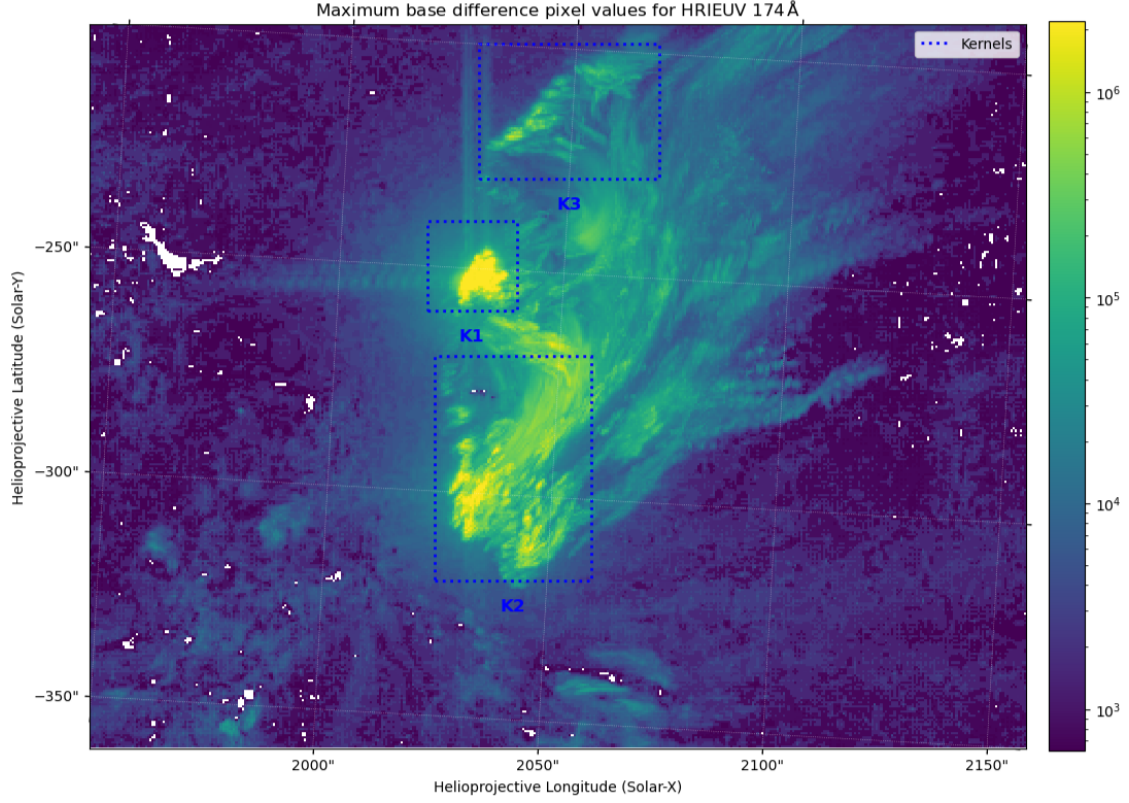
Furthermore, we show X-ray light curves from GOES and STIX, in multiple energy bands. STIX light curves display similar profiles to GOES light curves, especially in the lower energy bands. We can see time-varying signatures in the 15-25 keV (orange) energy band.



We compare Fermi GBM and STIX light curve for this specific energy band (15-25 keV). They look pretty similar, and that's good indicator for subsequent analysis. We applied time correction to STIX light curves in both of the last two cases.

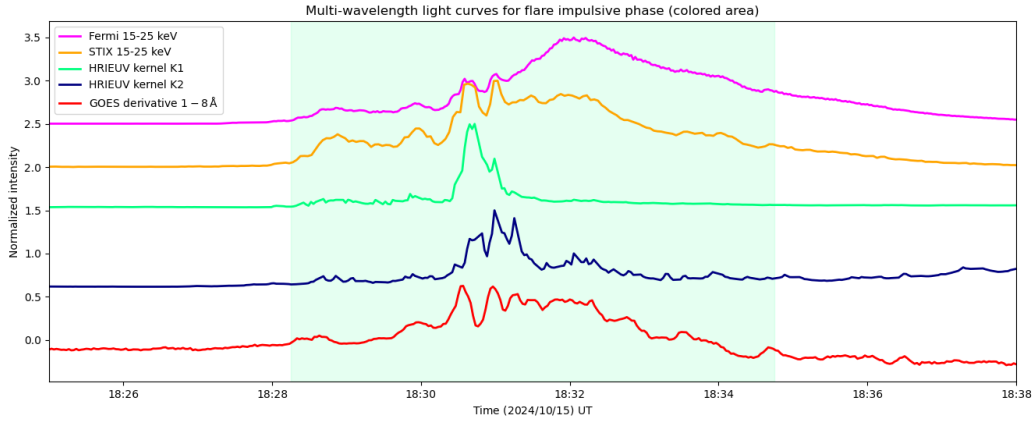


By examining HRIEUV maps for time range of interest, we identified few kernels to extract light curves from. We picked one reference map and called it the *base*. For multiple frames in the flare sequence, we subtracted the base image. This gave us a *difference image*, highlighting changes relative to the base. Then, instead of keeping all difference maps, we took the pixel-by-pixel maximum across the flare sequence. In other words, we built a map where the flare was brightest relative to the base. Then, we defined kernels and extracted light curves. Plots are on the next page.

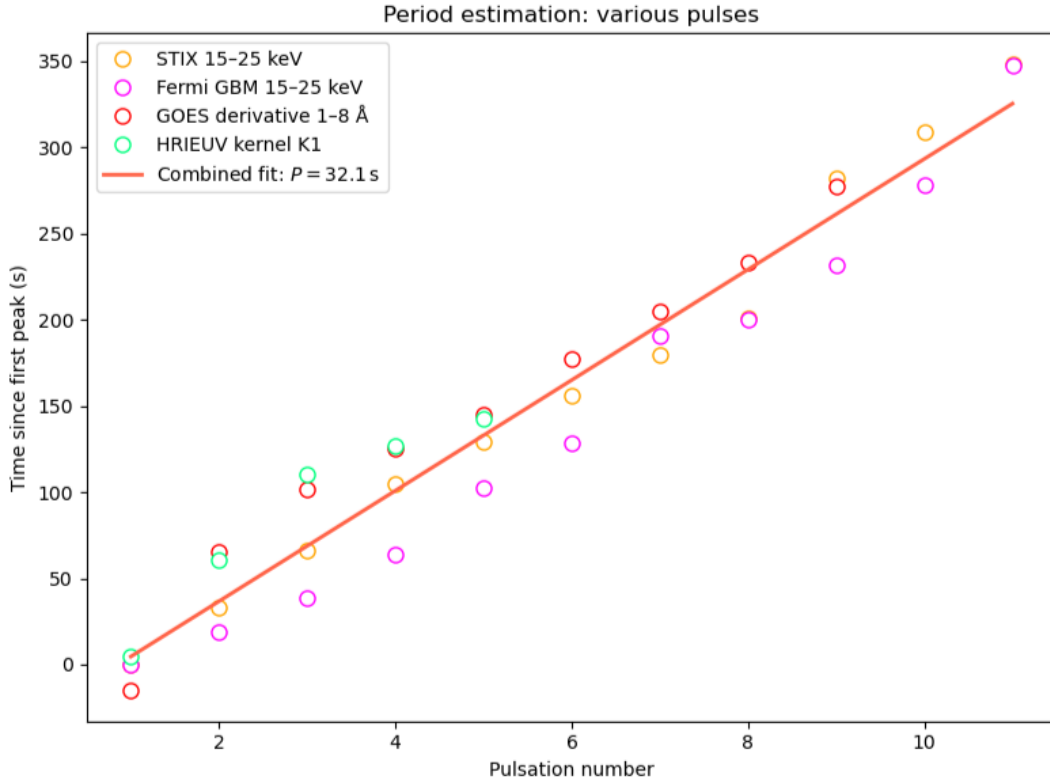


We could not confidently distinguish the ribbons from the filaments. However, kernel K1 is defined around a structure reminiscent of a footpoint, while kernels K2 and K3 cover regions that appear promising. Thus, their selection was not arbitrary.

We show overview of what we did so far on the next page.



Finally, we arrive at the stage of identifying pulsations. This was carried out using two complementary approaches: manual peak detection and the wavelet method.



Plot above gives us good approximation for period ($P = 32.1$ s). We aren't that confident about HRIEUV light curve², since we identified only 5 peaks - but we'll perform wavelet analysis to solidify our findings.

²Light curve extracted from kernel K1.

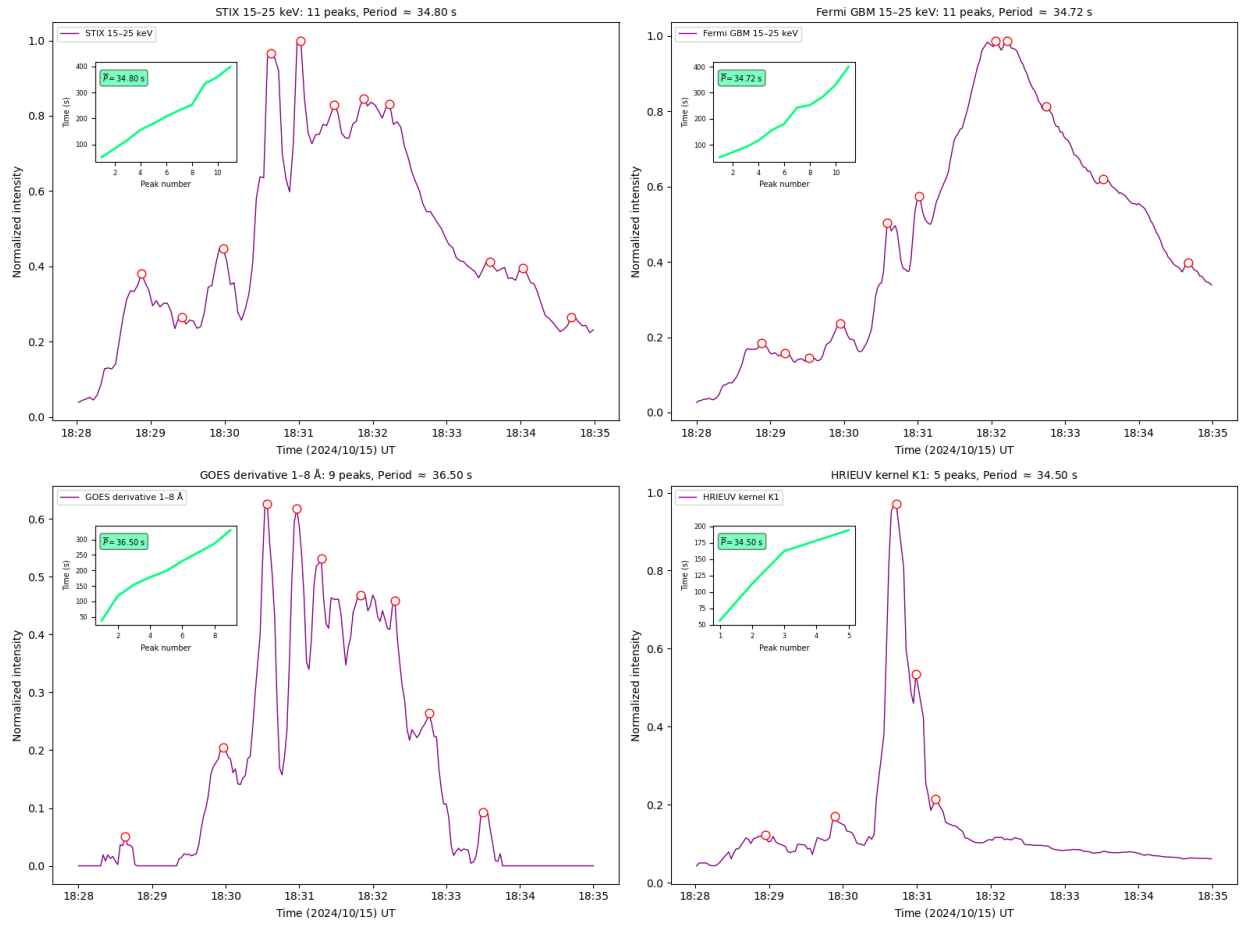
We also present another plot illustrating the manual peak-finding method, which allows us to estimate the individual period for each light curve.

STIX (15-25 keV) | $P = 34.80$ s

Fermi GBM (15-25 keV) | $P = 34.72$ s

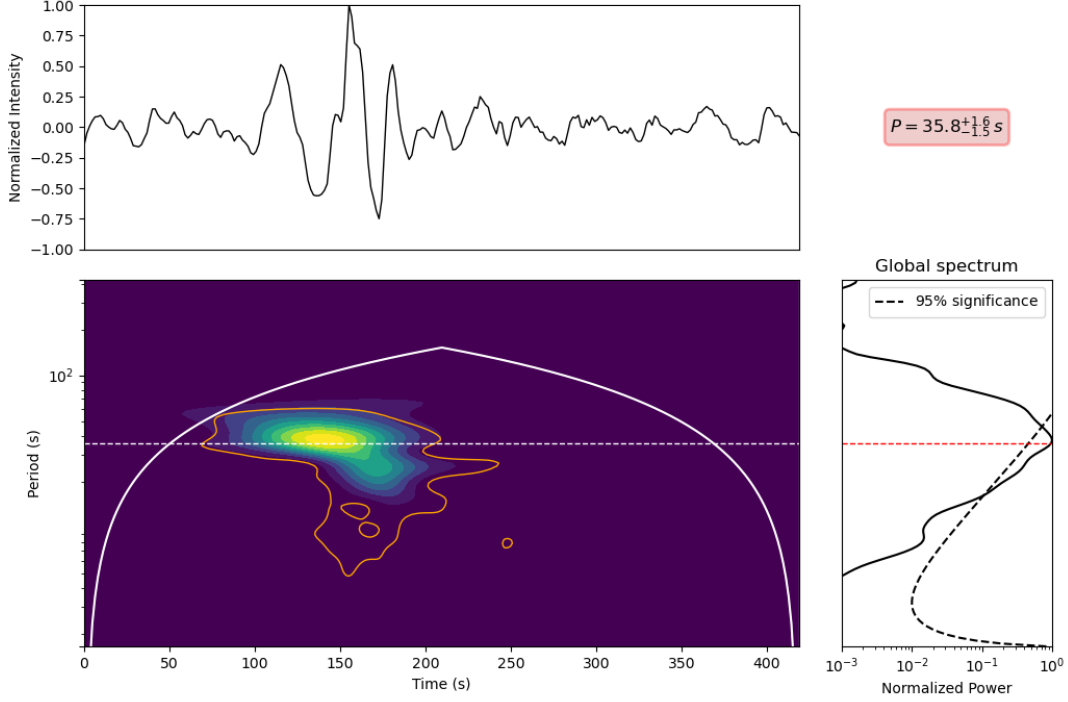
GOES derivative | $P = 36.50$ s

HRIEUV kernel K1 | $P = 34.50$ s



Some light curves were slightly smoothed as a part of preprocessing.

Next, we present the results of the wavelet analysis. The resulting plot for Fermi GBM (15–25, keV) is shown, where the analysis was performed with detrending included. This procedure was applied to all light curves.



Below, we present a table of the periods obtained through wavelet analysis. These results are in good agreement with those derived from the manual peak-finding method.

λ	STIX (15–25 keV)	Fermi (15–25 keV)	174 Å	1–8 Å
Period (s)	$36.6^{+1.6}_{-1.6}$	$35.8^{+1.6}_{-1.5}$	$39.3^{+7.4}_{-6.3}$	$36.6^{+3.3}_{-3.0}$

Additional remarks

1. The fact that we observe the same ~ 35 s period across hard X-ray (STIX and Fermi GBM), soft X-ray (GOES), and EUV (HRIEUV) strongly indicates few things: common electron population driving emissions across energy ranges, periodic acceleration/injection mechanism (rather than merely plasma oscillations) and coherent energy-release process affecting the entire flare structure.
2. (**highly speculative**) Looking at the flare, it seems that multiple reconnection sites play a role in driving the energy release. We speculate that periodic formation of magnetic islands in the current sheet, combined with stochastic bursts of energy shaped by the magnetic configuration, could explain the pulsations we see. Again, period we found hints stable magnetic structure and moderate spatial scale within the flare.
3. For future work, it would be valuable to track the spatial evolution of kernels K1, K2, and K3 (however, this kernel might not be that important) over time, since this could shed light on sequential activation, apparent motion along the ribbons, or changes in relative brightness. Understanding the magnetic context (filament eruptions or complex active region structures) would provide essential background, and investigating the phase relationships of pulsations across wavelengths could clarify whether they are in phase or exhibit time delays.